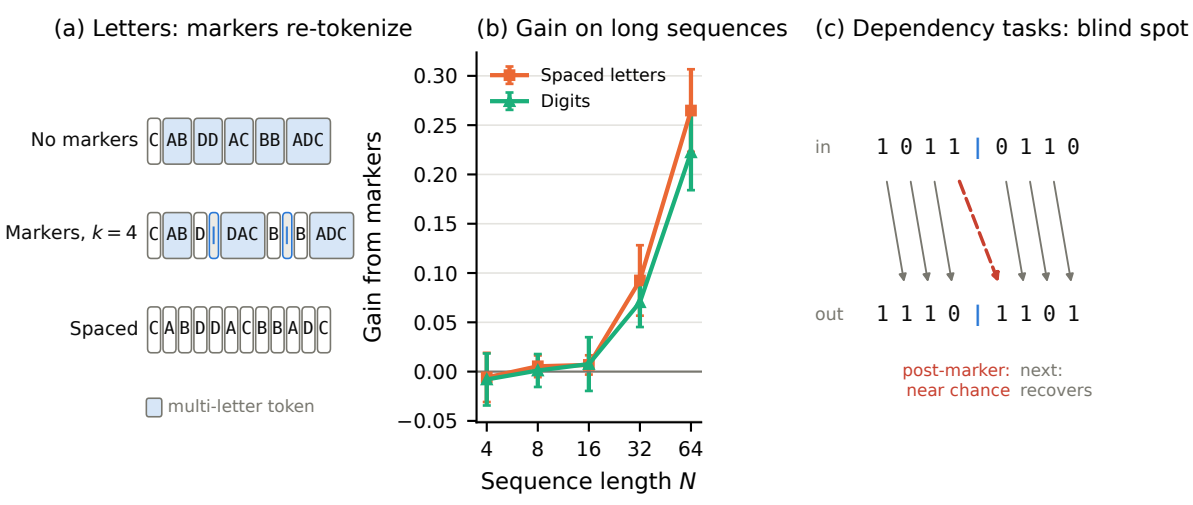
(a) Letters: markers re-tokenize
No markers
C
AB
DD
AC
BB
ADC
Markers, k = 4
C
AB
D
DAC
B
B
ADC
Spaced
C A B D D A C B B A D C
multi-letter token
(b) Gain on long sequences
Spaced letters
Digits
Gain from markers
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
4
8
16
32
64
Sequence length N
(c) Dependency tasks: blind spot
in
1 0 1 1 | 0 1 1 0
out
1 1 1 0 | 1 1 0 1
post-marker: near chance
next: recovers